System Of Linear Equations

A system of linear equations can be solved by

2 methods.



VOCABULARY

System : A set of 2 or more equations with 2 or more variables

$$x + y = 5$$
$$2x + 3y = 10$$

$$x + y + z = 1$$
$$2x - 3y + z = 5$$
$$x + y - 4z = 2$$

Linear : An equation with max degree 1 in x.

$$y = x + 5 \quad , \quad y = 2x - 6 \quad , \quad 5x + 2y = 7$$

Quadratic : An equation with max degree 2 in x.

$$y = x^2 \quad , \quad y = 5x^2 + 2 \quad , \quad y = x^2 + 3x - 7$$

Elimination Method

We eliminate a variable we choose. Let it be x.

$$x + 3y = 13$$
$$\ominus \quad x - y = 5$$
$$4y = 8$$
$$y = 2$$

Solve for x:

$$x - y = 5$$
$$x - 2 = 5$$
$$x = 7$$

Solution: $(7, 2)$

Elimination Method

We eliminate a variable we choose. Let it be y.

$$
\begin{array}{rrcrcl}
 & x & + & 3y & = & 13 \\
\oplus & [\,x & - & y & = & 5\,] \times 3 \\
\hline
 & 4x & & & = & 28
\end{array}
$$

$$x = 7$$

Solve for y &

$$x - y = 5$$
$$7 - y = 5$$
$$y = 2$$

Solution: $(7, 2)$

Both give the same answer!

Substitution Method

We solve for a variable then substitute

$$x + 3y = 13$$
$$x - y = 5$$

Method ①

Solve for x in ① &

$$x = 13 - 3y$$

Substitute in ②

$$x - y = 5$$
$$13 - 3y - y = 5$$
$$13 - 4y = 5$$
$$-4y = -8$$
$$y = 2$$

Solve for x :

$$x - y = 5$$
$$x - 2 = 5$$
$$x = 7$$

Solution : $(7, 2)$

Substitution Method

We solve for a variable then substitute.

$$x + 3y = 13$$
$$x - y = 5$$

Method ②

Solve for y in ②:

$$y = x - 5$$

Substitute in ①

$$x + 3y = 13$$
$$x + 3(x-5) = 13$$
$$4x - 15 = 13$$
$$4x = 28$$
$$x = 7$$

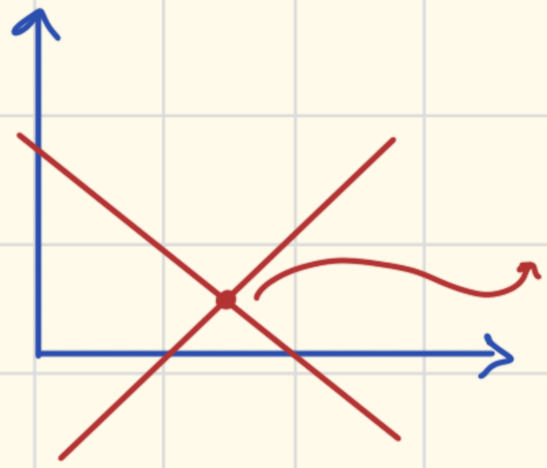
Solve for y &

$$x - y = 5$$
$$7 - y = 5$$
$$y = 2$$

Solution: $(7, 2)$

Solutions Of Systems

1) Only 1 solution



Solution:

only 1 intersection

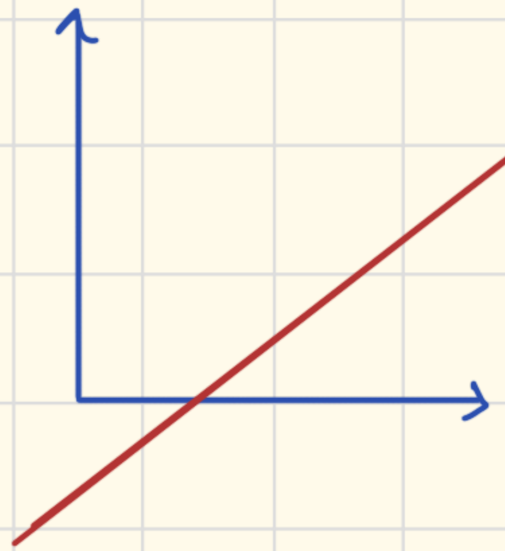
2) NO solutions



No intersection

No solution

3) INFINITE solutions



Two lines on each other

infinite solutions

Solutions Of Systems

1) Only 1 solution

2) NO solutions

$$\left.\begin{aligned} x + 5y &= 10 \\ x + 5y &= 15 \end{aligned}\right\} \quad 10 \neq 15$$

No solution

- Parallel lines

3) INFINITE solutions

$$\left.\begin{aligned} x + 5y &= 10 \\ 2x + 10y &= 20 \end{aligned}\right\} \quad 0 = 0$$

All points on $x + 5y = 10$

solutions

- Same lines

System Of Non-Linear Equations

Use substitution and quadratic solving techniques.

$$y = x^2 - 4x + 8$$
$$y = 2x$$

Substitute ② in ① 8

$$x^2 - 4x + 8 = 2x$$
$$x^2 - 6x + 8 = 0$$
$$(x-2)(x-4) = 0$$
$$x = 2 \qquad x = 4$$

$y = 2 \times 2 = 4$ $\quad (2, 4)$

$y = 2 \times 4 = 8$ $\quad (4, 8)$